Pratheba Selvaraju

3D Vision & Graphics · Geometry Processing · Generative 3D

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Profile

Research scientist and engineer working at the intersection of **3D computer vision, computer graphics, and geometry processing**. Research spans structured 3D representations, reconstruction from imperfect real-world observations, generative modeling, shape deformation, and production-oriented geometric systems.

Experience

Senior Research Scientist, *SmileShape.ai* June 2025 – Present
Geometry Processing & 3D Computer Graphics

- Develop and productionize high-impact geometry processing and 3D computer graphics components for complex 3D modeling workflows.
- Design robust geometric algorithms for mesh processing, shape construction, deformation, and topology-sensitive operations, with emphasis on production reliability.
- Drive components from mathematical formulation and prototyping through diagnosis, validation, and integration into production systems.

Research Scientist, *Max Planck Institute for Intelligent Systems – Perceiving Systems* Feb. 2025 – Present
Tübingen, Germany · directed by Dr. Michael J. Black

- Research 3D shape understanding, reconstruction, and deformation, with emphasis on generative and geometry-aware representations.
- Work on sewing-pattern estimation, digital human modeling, and methods that connect learned 3D representations with explicit geometric structure.

Research Intern, *Roblox* 2024
California

- Developed geometry-processing methods for adapting template garments across topologically and morphologically different Roblox avatars.
- Investigated implicit surface reconstruction for robust shape deformation and transfer.

Research Intern, *Microsoft* 2022
Washington (Remote)

- Developed diffusion-based methods for 3D face reconstruction from single-view images.

Software Engineer Intern, *Google* 2022
California (Remote)

- Worked on 3D semantic understanding for identifying building components from LiDAR point clouds and imagery using implicit models and point-cloud segmentation.

Research Intern, *Meta* 2020
Washington (Remote)

- Worked on virtual object placement for AR/VR environments, including placement of virtual display surfaces in Oculus scenes.

Software Engineer, *IMO* California, 2017
Software Engineer, *Machine Zone* California, 2016
Software Engineer, *Microsoft* Washington, 2013–2016

Education

Ph.D., Computer Science – *University of Massachusetts Amherst* 2018 – 2024
Thesis: Exploring Representations for 3D Reconstruction from Impaired Real-World Data · Advisor: Prof. Erik G. Learned-Miller
M.S., Computer Science – *Columbia University* 2011 – 2012

Selected Publications

- **NGL-Prompter: Training-free Sewing Pattern Estimation from Images** Under review, 2026
  Anna Badalyan*†, **Pratheba Selvaraju**†, Victoria Fernández Abrevaya, Omid Taheri, Michael Black
- **OFER: Occluded Face Expression Reconstruction** **CVPR 2025**
  Pratheba Selvaraju, Victoria Fernández Abrevaya, Timo Bolkart, Rick Akkerman, Tianyu Ding, Faezeh Amjadi, Ilya Zharkov
- **FORA: Fast-Forward Caching in Diffusion Transformer Acceleration** arXiv, 2024
  Pratheba Selvaraju, Tianyu Ding, Tianyi Chen, Ilya Zharkov, Luming Liang

- **Developable Approximation of Neural Implicits via Rank Minimization** **3DV 2024**
  Pratheba Selvaraju
- **A 3D digitisation workflow for architecture-specific annotation of built heritage** JASREC, 2021
  Marissia Deligiorgi, Maria I. Maslioukova, Melinos Averkiou, Andreas C. Andreou, **Pratheba Selvaraju**, Evangelos Kalogerakis, Gustavo Patow, Yiorgos Chrysanthou, George Artopoulos
- **BuildingNet: Learning to Label 3D Buildings** **ICCV 2021 (Oral)**
  Pratheba Selvaraju, Mohamed Nabail, Evangelos Kalogerakis, Siddhartha Chaudhuri

Selected Research Projects

Shape Deformation with Style Preservation *First Author; in preparation for submission*
Adaptation of 3D accessories and fine geometric details to topologically and morphologically different shapes for gaming and animation.

VOFER: Video & Audio-Based Occluded Face Expression Reconstruction *Corresponding Author*
Extension of OFER to multimodal reconstruction from synchronized audio and video sequences.

Research Service & Mentoring

Reviewer: CVPR · ICCV · ECCV · SIGGRAPH · SIGGRAPH Asia · 3DV · TPAMI · TVCG · IJCV

Student mentoring: Dharmendra Selvaratnam, University of Plymouth, London

Technical Expertise

Languages	Python, C++
ML / Scientific	PyTorch, NumPy, SciPy; diffusion models, transformers
3D / Geometry	Geometry processing, mesh processing, implicit representations, point clouds, shape deformation, 3D reconstruction
Research Areas	3D computer vision, 3D computer graphics, generative 3D, digital humans, novel-view synthesis